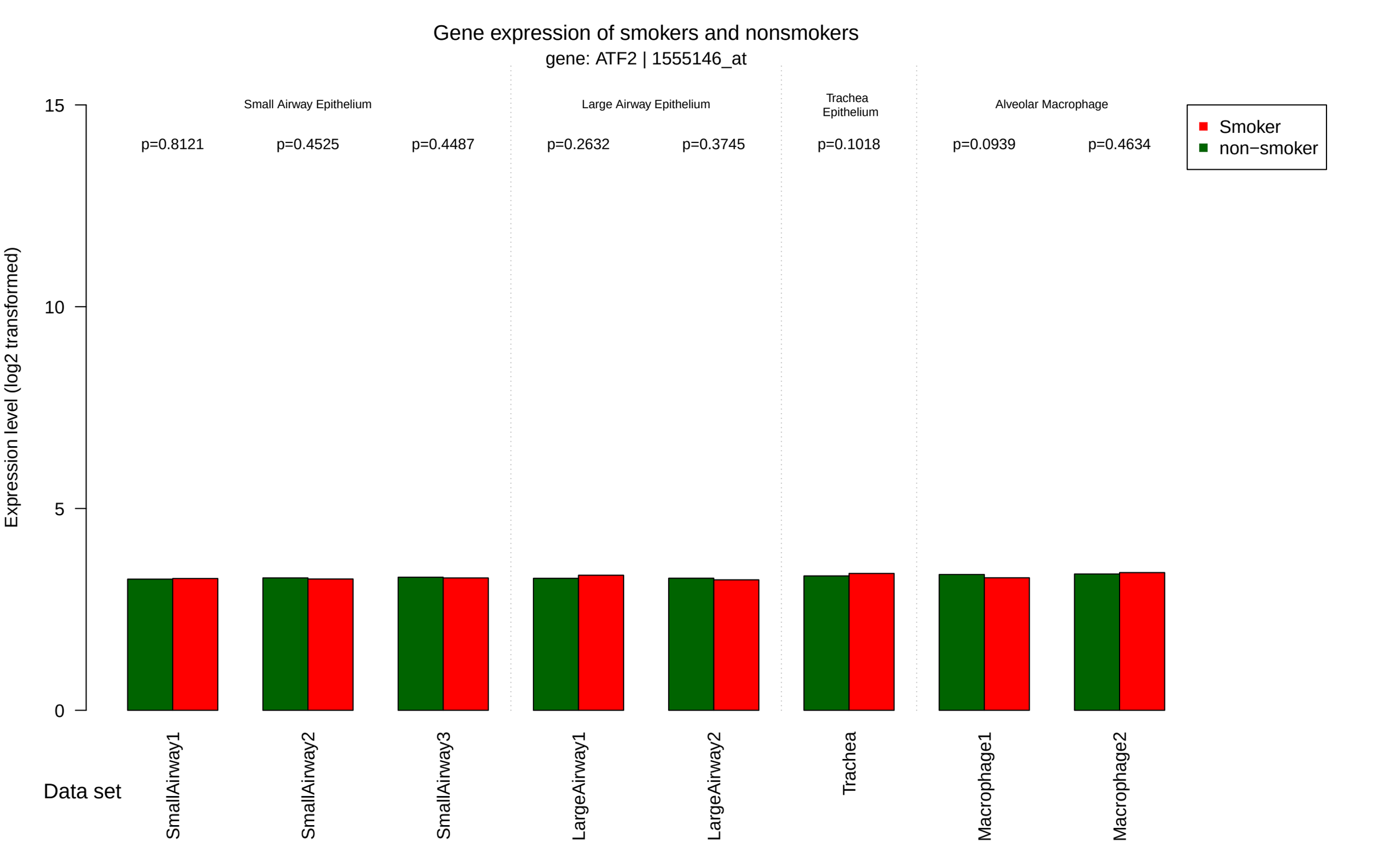
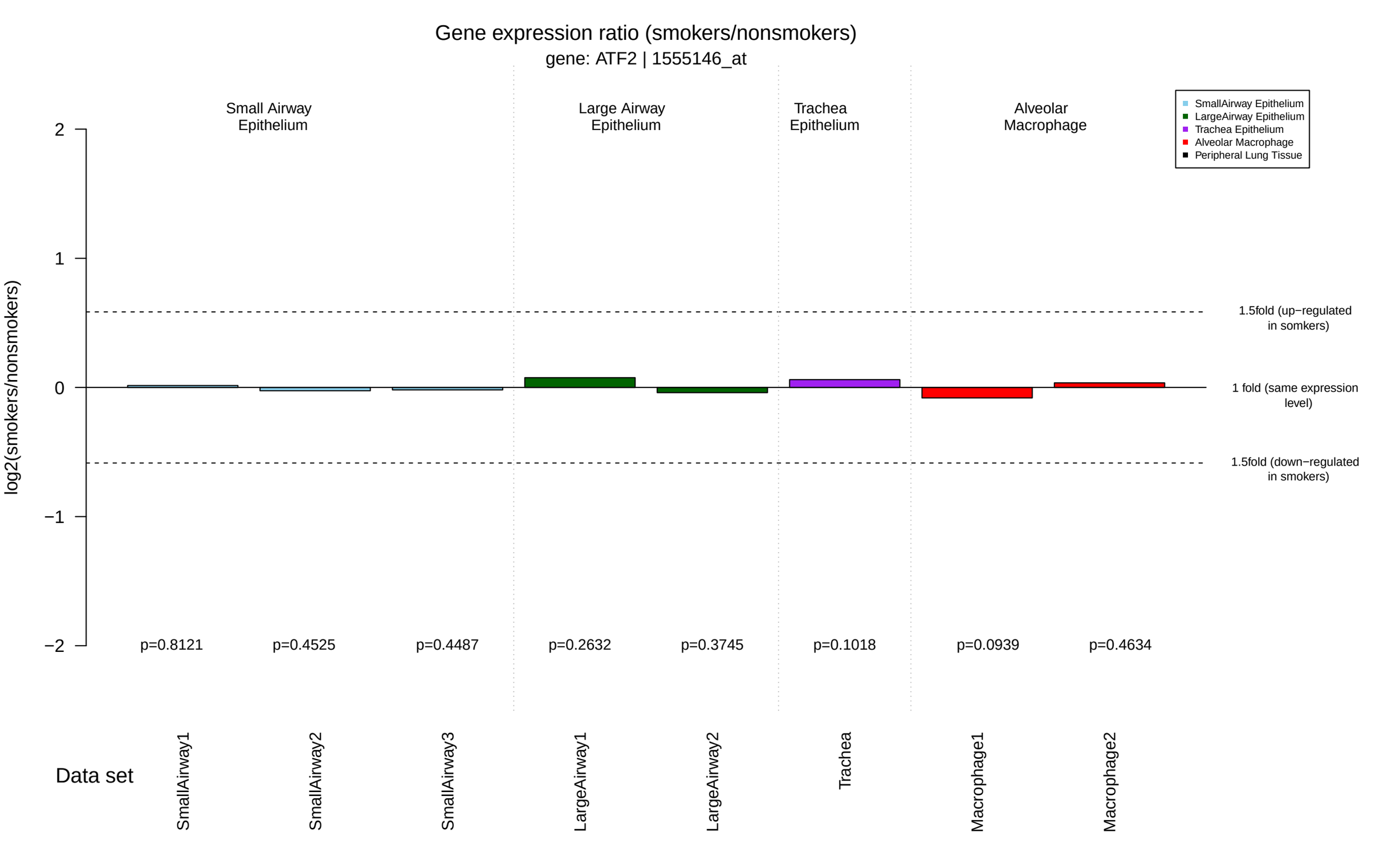
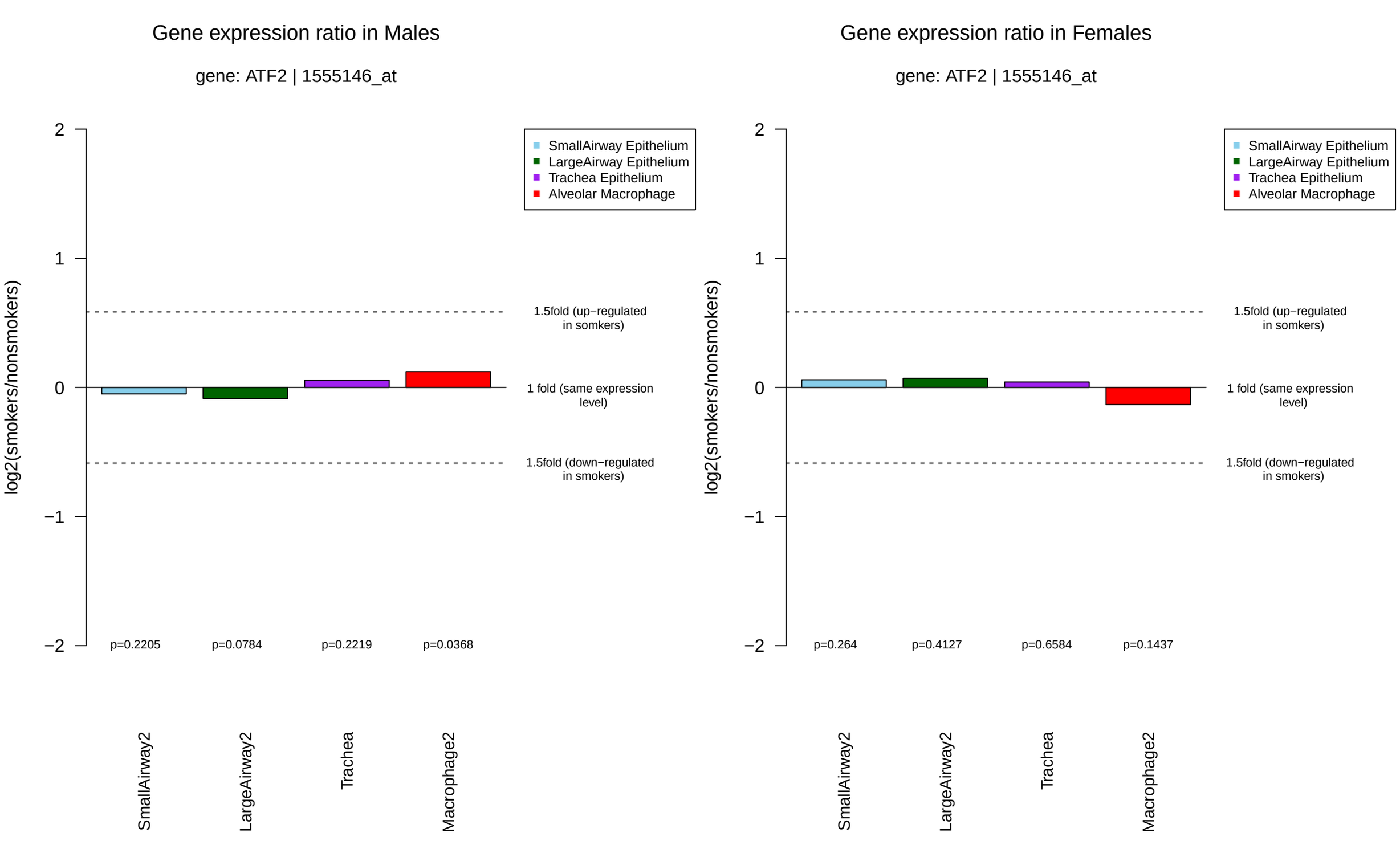
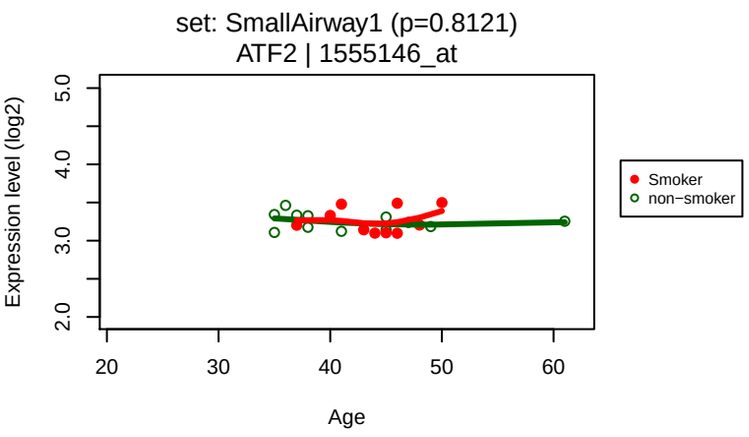
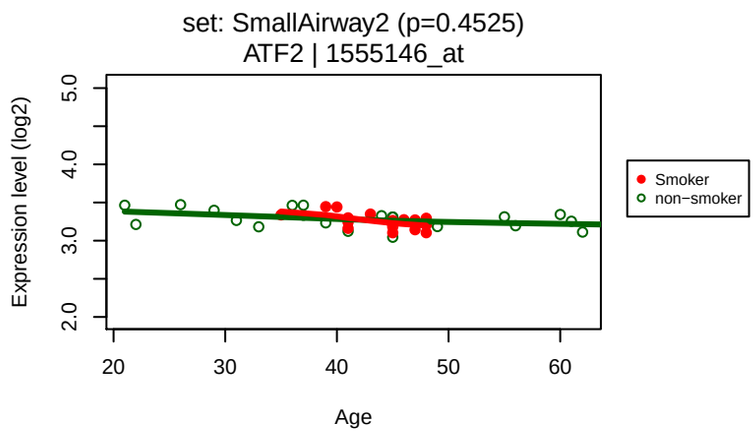
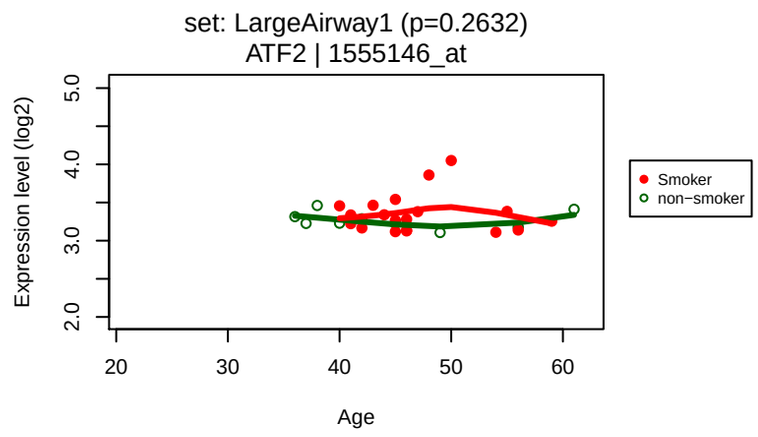

Gene expression of smokers and nonsmokers
gene: ATF2 | 1555146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8121
p=0.4525
p=0.4487
p=0.2632
p=0.3745
p=0.1018
p=0.0939
p=0.4634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

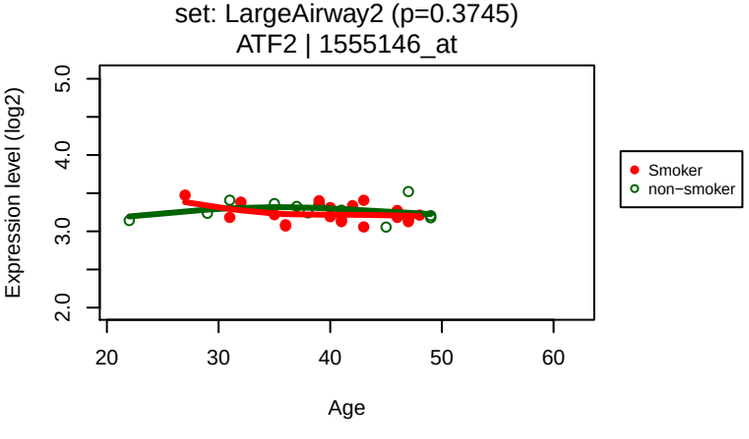

Gene expression ratio (smokers/nonsmokers)
gene: ATF2 | 1555146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8121
p=0.4525
p=0.4487
p=0.2632
p=0.3745
p=0.1018
p=0.0939
p=0.4634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

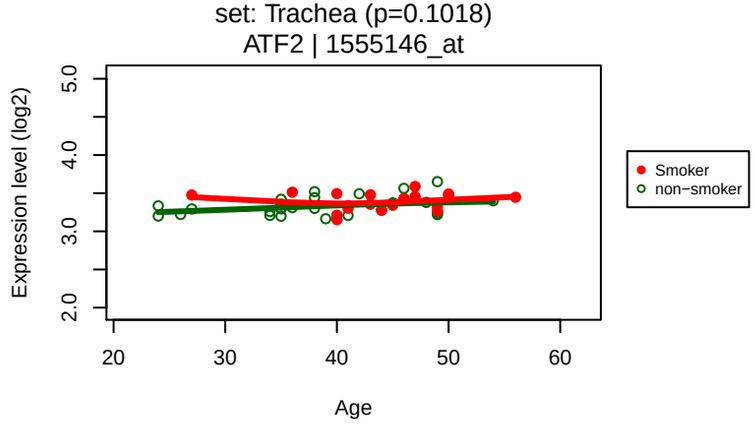

Gene expression ratio in Males
gene: ATF2 | 1555146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2205
p=0.0784
p=0.2219
p=0.0368
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF2 | 1555146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.264
p=0.4127
p=0.6584
p=0.1437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

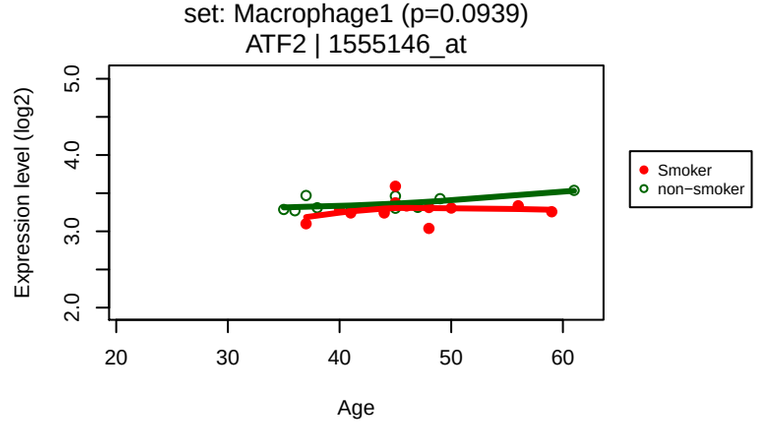

set: SmallAirway1 (p=0.8121)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker

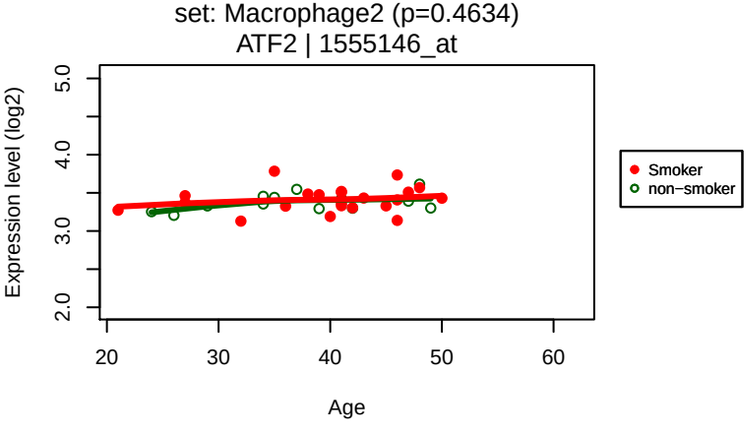

set: SmallAirway2 (p=0.4525)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker

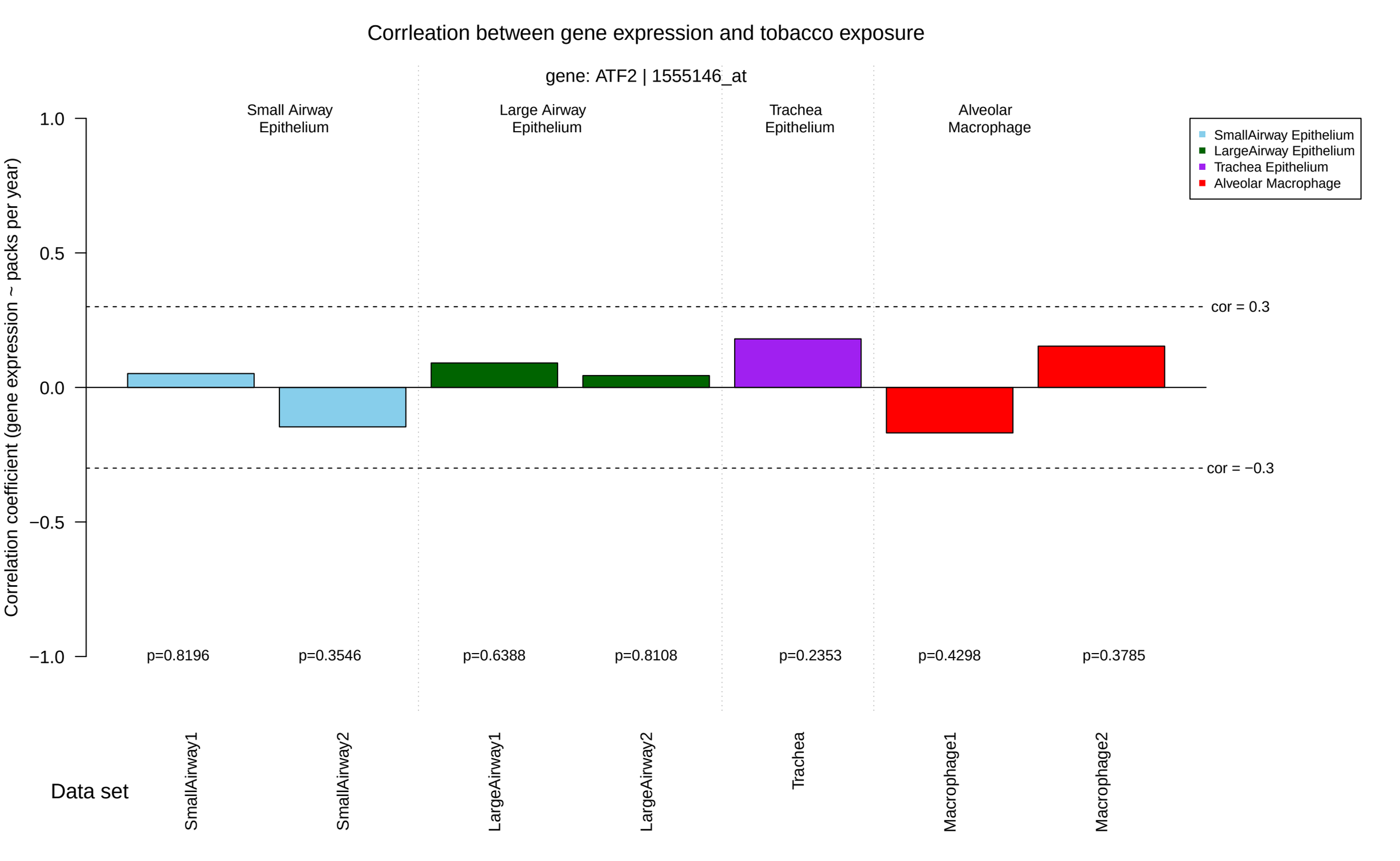

set: LargeAirway1 (p=0.2632)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3745)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1018)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0939)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4634)
ATF2 | 1555146_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF2 | 1555146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8196
p=0.3546
p=0.6388
p=0.8108
p=0.2353
p=0.4298
p=0.3785
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2